

In Maclaurin a is always $= 0$

The rest is the same as Taylor series.

o) e^x

when $x=a$ $\therefore$ which is always 0

$$f(x) = e^x = 1$$

$$f'(x) = e^x = 1$$

$$f''(x) = e^x = 1$$

$$f'''(x) = e^x = 1$$

$$f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!}$$

$$= 1 + \frac{1(x-0)}{1!} + \frac{1(x-0)^2}{2!} + \frac{1(x-0)^3}{3!}$$

$$= 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$= \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

②) $\sin x$

$$x \in a \quad a = 0$$

$$f(a) = \sin x = 0$$

$$f'(a) = \cos x = 1$$

$$f''(a) = -\sin x = 0$$

$$f'''(a) = -\cos x = -1$$

$$f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!}$$

$$= 0 + \frac{1(x)}{1!} + 0 - \frac{x^3}{3!}$$

$$= x - \frac{x^3}{3!} \dots \dots$$

③) $\frac{1}{1-x}$

$$x \in a \quad a = 0$$

$$f(a) = \frac{1}{1-x} = 1$$

$$f'(a) = \frac{1}{(1-x)^2} = 1$$

$$f''(a) = \frac{2}{(1-x)^3} = 2$$

$$f'''(a) = \frac{6}{(1-x)^4} = 6$$

$$f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!}$$

$$= 1 + \frac{x}{1!} + \frac{2x^2}{2!} + \frac{6x^3}{3!}$$

$$\therefore (1-x)^{-1}$$

$$= -1(1-x)^{-2}(-1)$$

$$= \frac{1}{(1-x)^2}$$

$$\therefore (1-x)^{-2}$$

$$= -2(1-x)^{-3}(-1)$$

$$= \frac{2}{(1-x)^3}$$

$$\therefore = 2(1-x)^{-3}$$

$$= 2(-1)(1-x)^{-4}(-1)$$

$$= \frac{6}{(1-x)^4}$$

